Higher National Diploma in Engineering (E/M/C/BSE)

First Year, First Semester Examination - 2018

MA 1101: Engineering Mathematics

Time:2 Hours

Question - 01 (Total 25 Marks)

a) Find the characteristic equation of the matrix

$$A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}$$

and Verify that the matrix **satisfies its own characteristics equation**. (12 Marks)

b) Find A^{-1} by using **Cayley - Hamilton Theorem**

$$A = \begin{pmatrix} 5 & 1 & 2 \\ 3 & -1 & 4 \\ -3 & 2 & 1 \end{pmatrix}$$

(08 Marks)

c) Write down the **quadratic form** corresponding to the matrix. (05 Marks)

Question - 02 (Total 25 Marks)

Test for **convergence** the series

a)

$$\frac{1}{2\sqrt{1}}+\frac{x^2}{3\sqrt{2}}+\frac{x^4}{4\sqrt{3}}+\frac{x^6}{5\sqrt{4}}\cdots$$

(15 Marks)

b)

$$\frac{1}{4.7.10},\frac{4}{7.10.13},\frac{9}{10.13.16},\cdots$$

(10 Marks)

Question - 03 (Total 25 Marks)

a) If $V=(x^2+y^2+z^2)^{\frac{m}{2}}$ then find the value of m ($m\neq 0$) which will make

$$\frac{\partial^{2}v}{\partial x^{2}}+\frac{\partial^{2}v}{{\partial y^{2}}+\frac{\partial^{2}v}{\partial z^{2}}=0$$

(10 Marks)

b) If $y=\sin(\sin x)$ Prove that $\frac{d^2y}{dx^2}+\tan x\frac{dy}{dx}+y\ \cos^2 x=0$ (08 Marks)

c) If $u=\sec^{-1}(\frac{x^3-y^3}{x+y})$ Using **Euler's theorem on homogeneous function**, Show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=2\ \cot\ u$ (07 Marks)

Question - 04 (Total 25 Marks)

a) If $U = xyz$, $V = x^2 + y^2 + z^2$, $W = x + y + z$ Find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ and $\frac{\partial(x,y,z)}{\partial(u,v,w)}$ (15 Marks)

b) i) Evaluate

$$\lim_{x \to \infty} \left(\frac{2xy + 3}{3x^2 + y^2} \right)$$

(05 Marks)

ii) Evaluate

$$\lim_{x \to 0} \frac{\log(1 + 2x)}{\sin\ 2x}$$

(05 Marks)

Question - 05 (Total 25 Marks)

a) Evaluate the **double integral** $\iint_D xy\, dx\, dy$ Where D is the region bounded by the x-axis the line $y = 2x$ and the parabola $x^2 = 4ay$. (10 Marks)

b) Calculate the **volume** of the solid bounded by the following surfaces $z = 0$, $x^2 + y^2 = 1$ and $x + y + z = 3$ (15 Marks)